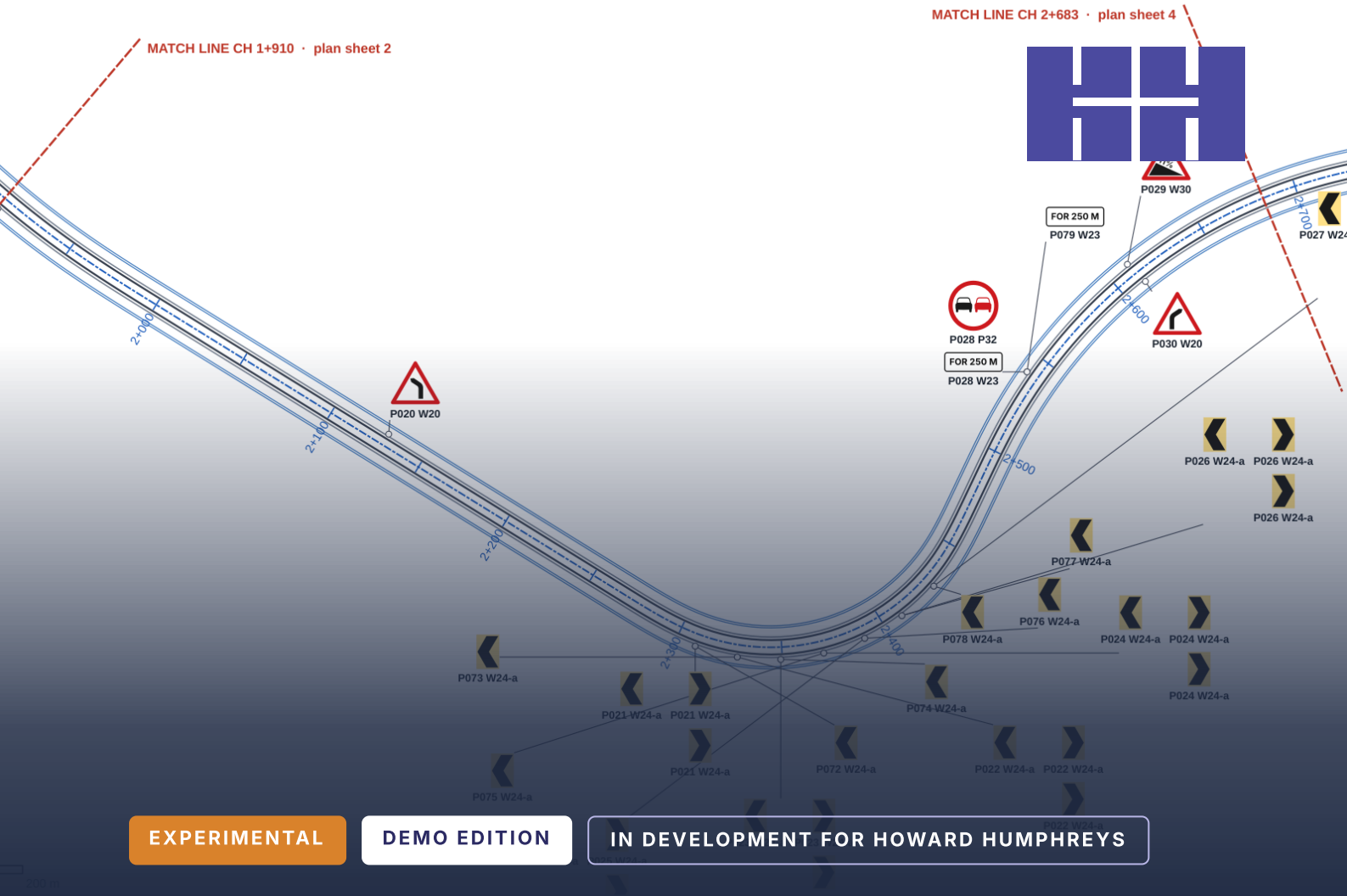


EXPERIMENTAL · DEMO EDITION · IN DEVELOPMENT FOR HOWARD HUMPHREYS

QGIS PLUGIN · VERSION 0.6.1

RoadSign Designer

User Guide and Technical Manual

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Developed for	Howard Humphreys (East Africa) Ltd, Roads and Transportation Department

Status	Experimental, in development
QGIS	3.44 LTR (tests run headless on 3.34)
Civil 3D	2026 and later, by copy and paste from a georeferenced DXF
Standard	Road Design Manual Part 6.2, Traffic Signs (Ministry of Roads and Transport, 2025)
Demo data	Olkaria LOT 3 design alignment (Civil 3D LandXML, Arc 1960 / UTM 37S)
Edition	November 2026

Internal working document. RoadSign Designer is under development for Howard Humphreys (East Africa) Ltd; behaviour, defaults and outputs may change between versions. Check every sign against RDM 6.2 and the project's design criteria before issue.

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PART 1

Getting started

What RoadSign Designer is for, what this edition can and cannot do yet, and how to get from a Civil 3D LandXML to a signed road in a few minutes.

1 About this edition

EXPERIMENTAL · DEMO EDITION · IN DEVELOPMENT FOR HH

This guide describes version 0.6.1, an experimental build prepared to demonstrate the workflow. Every figure comes from the Olkaria LOT 3 design alignment, used here as demonstration data. The tool is not yet released for project use: treat its proposals as a first draft for an engineer to review, not as a design.

What has been checked so far:

- The core library has 37 automated tests and the QGIS plugin 27 headless tests (QGIS 3.34, offscreen).
- Alignment geometry was checked against two real Civil 3D exports: Olkaria LOT 3 (UTM 37S, 59 MB with TIN surfaces and 3,569 corridor sections) and the Nessuit–Salgaa preliminary design (UTM 36S, four alignments, 26 spirals).
- DXF georeferencing was checked on Olkaria: the reference alignment starts exactly on the LandXML's first point and every exported sign maps back to its chainage and offset with no measurable difference (section 24).
- The copy-and-paste route was tried in Civil 3D: signs copied from the Olkaria DXF with COPYBASE and PASTEORIG landed on the alignment in the project drawing.

What has **not** been checked yet:

- The auto-placement rules follow RDM 6.2, but the sight-distance and side-friction values are AASHTO 2011 defaults until they are checked against RDM Part 1.3 (section 21).

2 What RoadSign Designer does

Placing traffic signs in Civil 3D and then counting them for the schedules is slow, manual work. RoadSign Designer moves both jobs into QGIS, next to the road alignment:

1 · LandXML	2 · Place	3 · Inventory	4 · Report	5 · Civil 3D
Alignment, profile, speeds, superelevation and corridor edges from Civil 3D.	Click signs on the map, or let the engine propose them from RDM 6.2.	Chainage, offset, side, facing and post, kept live as signs move.	Plan and long-section sheets, schedule and findings (PDF, PNG, Excel).	Georeferenced DXF, pasted into the project drawing at the true coordinates.

Each sign is a point in a GeoPackage. Its chainage, offset, side, facing, size, grade and clearance from the road edge are worked out from the alignment and recomputed whenever the sign is moved. The inventory and the checks update as you work.

Scope:

- **Placement first.** The tool does not extract signs from existing drawings, because many drawings show signs as images rather than blocks.
- **Inventory, not a Bill.** Outputs are counts and chainages (Excel Tables and CSV) that feed the separate QTO and Bill process. There are no rates.
- **QGIS front end.** Signs are placed on a map, not typed into a spreadsheet.
- **Civil 3D 2026+ through a georeferenced DXF.** No compiled .NET add-in.

3 Installing the plugin

1. In QGIS, open **Plugins → Manage and Install Plugins → Install from ZIP** and choose `roadsign_designer.zip`.
2. Enable **RoadSign Designer**. A dock opens with the **Place**, **Inventory**, **Auto** and **Signs** tabs; the toolbar button reopens it.
3. Nothing else needs installing. The libraries used for exports (openpyxl, ezdxf and their dependencies) are pure Python and ship inside the plugin's `ext_libs` folder, so the plugin works where `pip` is blocked.

> **UPDATING**
> Install the new ZIP over the old one and restart QGIS. Existing sign projects open in newer versions; missing fields are added automatically.

4 Quick start

1. **New project...**: choose the LandXML, the alignment and where to save the GeoPackage. The layer CRS defaults to the LandXML's (EPSG:21037 for Olkaria).
2. **Auto tab → Propose signs**. Proposed signs appear with an orange ring. Look through them, then **Accept all proposals** or remove the ones you do not want.
3. **Place tab**: add anything the engine does not cover (junction and direction signs, place names) by picking a sign and clicking on the map.
4. **Inventory tab → Road signage report**: choose the plan scale and fill in Prepared / Checked / Approved by.

5. **Inventory tab → Export to Civil 3D**: choose the plan scale, then paste the signs into the project drawing (section 16).

PART 2

Working in QGIS

Creating a sign project, placing and editing signs, writing legends, and the checks that run as you work.

5 A sign project from a LandXML

New project… asks for the LandXML, the alignment to sign (a file can hold several), and a GeoPackage to save. The GeoPackage is the master record: it holds the signs and a copy of everything derived from the LandXML that the map needs.

Layers added to the map:

Layer	Contents
RSD signs	One point per sign face, drawn with its RDM artwork (faces on one post are stacked).
RSD alignment	The signed alignment.
RSD chainage	Ticks and labels every 100 m.
RSD road edges	Corridor edges from the LandXML cross-sections, coloured by code (ETW, EPS, Hinge, Ditch_In, Ditch_Out, Daylight).
RSD vertical curves	Crest and sag PVIs with length, K and A.

The sign layer takes the LandXML's coordinate system when the file declares one. If the QGIS project uses a different CRS the plugin says so; QGIS reprojects on the fly, but keeping both the same avoids confusion when exchanging with Civil 3D.

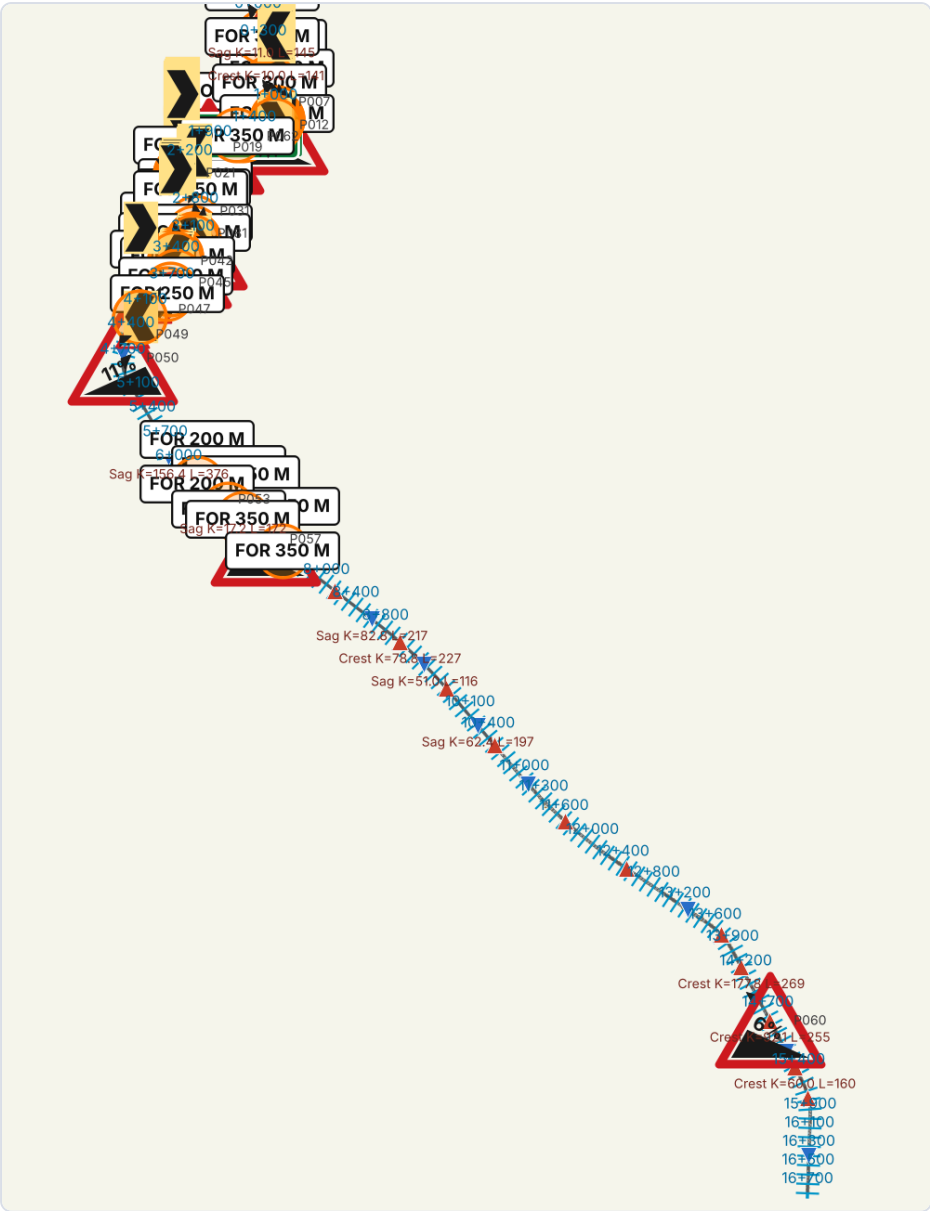


Figure 1. Olkaria LOT 3 after auto-placement and acceptance: the whole 16.7 km alignment with its signs.

6 Placing signs

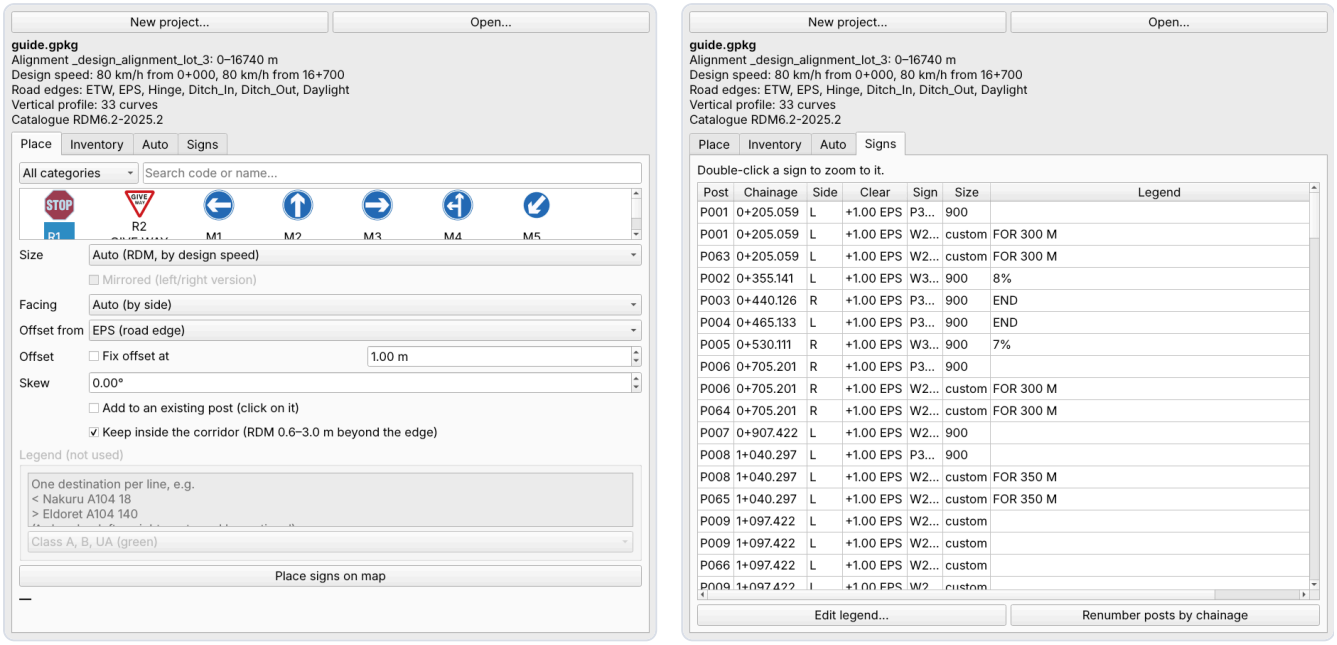


Figure 2. The Place tab (left): catalogue palette, size, facing, offset rule and legend. The Signs tab (right): every face in chainage order with its clearance from the road edge.

1. **Pick a sign** from the palette. Filter by category or search by code or name.
2. **Size.** *Auto (RDM, by design speed)* applies the manual's size tables using the design speed at the sign's chainage. Choose a size to override it.
3. **Mirrored** gives the left/right version of signs the manual shows one way round (bends, side roads, chevrons).
4. **Facing.** *Auto* follows driving on the left: a sign on the left faces up-chainage traffic, a sign on the right faces down-chainage traffic. Choose Forward or Backward to override, for example in a median.
5. **Offset from** a road edge (EPS by default) and **Fix offset at** place the post at a set distance beyond that edge at the chainage you click. The live readout shows, for example, `+1.00 m from EPS grade +2.5%`.
6. **Keep inside the corridor** (on by default) snaps the post into RDM 6.2 §2.4's band, 0.6 to 3.0 m beyond the edge, and never beyond the ditch or daylight line where those are modelled.
7. **Add to an existing post**: click an existing sign to put another face (a plate, a back-to-back sign) on the same post.
8. **Place signs on map**. Hovering shows chainage and offset; left-click places, right-click stops.

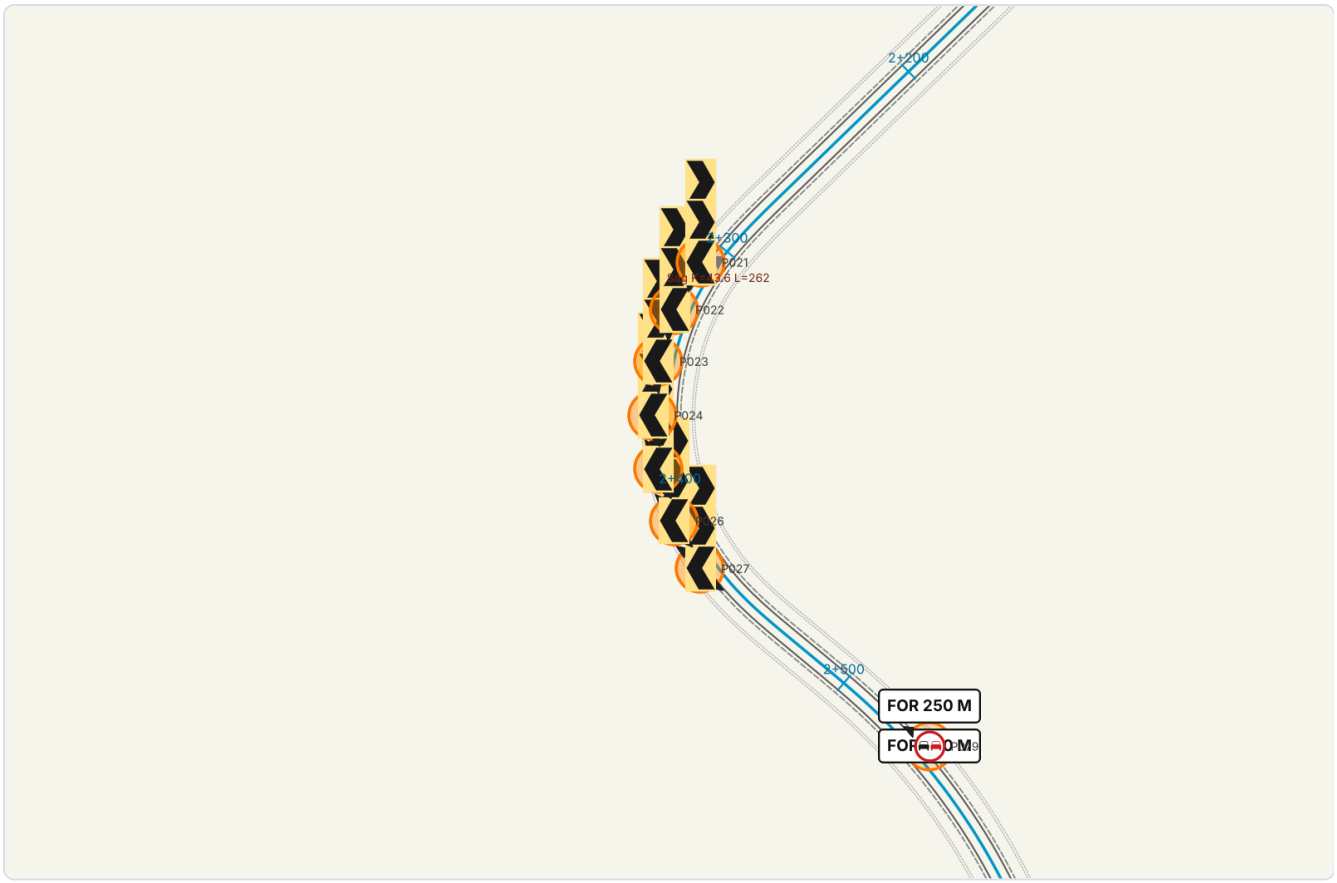


Figure 3. The QGIS map at the Olkaria chevron curve near CH 2+380: corridor edges, chainages and signs placed by the engine.

7 Legends, plates and values

Directional signs, town and county signs and text plates carry text that changes from job to job: a sign at Salgaa reads Nakuru and Eldoret, not Mombasa. Type the legend before placing, or select the sign on the Signs tab and use **Edit legend…**. One destination per line:

```
^ Olkaria 6              ahead, 6 km
< Naivasha B3 32         left, route B3, 32 km
> Narok B3 120           right
```

- Arrows: `^` or `A` ahead, `<` or `L` left, `>` or `R` right. A line with no arrow is plain text (plates).
- Destinations are ordered as RDM §5.2 asks: ahead, then left, then right, furthest first.
- Faces are drawn in the RDM Table 2.6 colours for the road class: green with yellow route numbers (A, B, UA), white (C, D, UC and lower), blue (expressway).
- Signs whose artwork carries a value take it from the legend. A W30 steep-descent sign proposed for an 11 % grade reads 11 %, not the manual's 10 % sample, on the map, in the report and in the DXF.



Figure 4. Legend faces drawn by the plugin: directional signs in three colour schemes, a distance plate and a place-name sign.

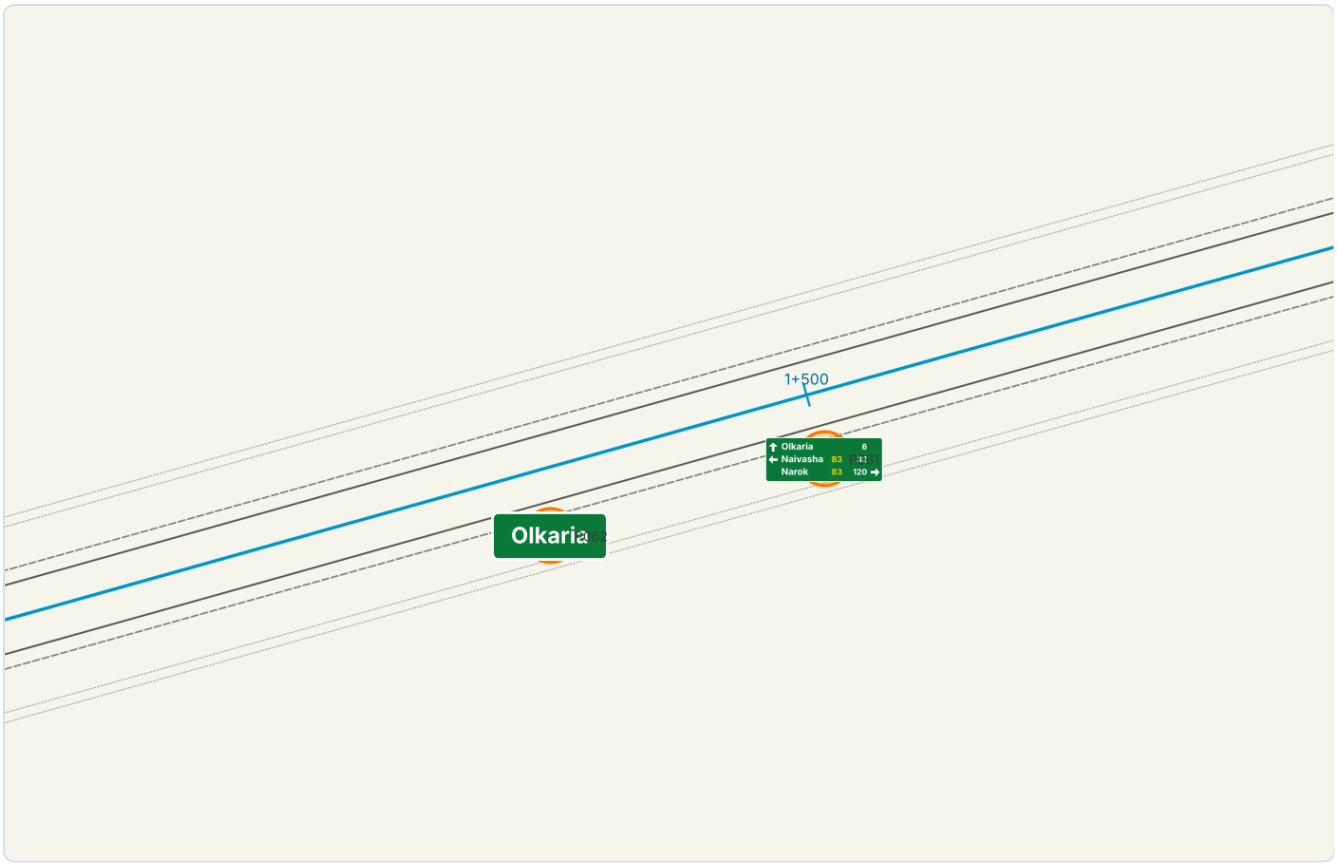


Figure 5. A directional sign and a place-name sign added by hand at Olkaria CH 1+500.

8 Moving and editing signs

- Move signs with the normal QGIS editing tools (Toggle Editing → Move Feature). Chainage, offset, side, auto-facing, size, grade and edge clearance update at once, and every face on the same post moves with it.
- Editing `facing` in the attribute table switches Auto facing off for that sign; typing a size switches Auto size off.
- The **Signs** tab lists every face in chainage order. Double-click a row to zoom to it. **Renumber posts by chainage** renames posts P001, P002… in road order.

9 Checks

The Inventory tab lists checks for the whole sign list. Errors block the Civil 3D export; warnings do not.

Check	Level	Rule
Inside the travelled way	Error	Post closer to the centreline than the edge of travelled way (ETW).
On the shoulder	Warning	Post between ETW and the edge of paved shoulder (EPS).
Outside the RDM lateral band	Warning	RDM 6.2 §2.4: 0.6 to 3.0 m beyond the carriageway or paved-shoulder edge.
Outside the corridor	Warning	Beyond the ditch or daylight line, where the verge is modelled.

Check	Level	Rule
Spacing	Warning	RDM 6.2 §2.4: consecutive signs at least 0.6 V metres apart (48 m at 80 km/h).
Legend missing	Warning	Directional signs and plates need their text.
Post consistency	Error	Faces on one post at different positions, or with different post or foundation types.
Unknown code, size, post or foundation	Error	Not in the catalogue.

PART 3

Auto-placement

The engine reads the alignment's geometry and proposes the warning and regulatory signs RDM 6.2 calls for. Proposals are suggestions: an engineer accepts or removes each one.

10 What the engine signs

Situation	Signs proposed	Basis
Curve whose safe speed is 10 km/h or more below the approach speed	W20 bend (W19 sharp turn when 30 km/h or more below), both directions, at the advance distance	RDM 6.2 §4.5; curve speed from $V^2 = 127\ R\ (e + f)$
Warned curves closer than 150 m	W22 series of bends (W21 sharp) with a W23 distance plate	RDM 6.2 §4.5
Curve 20 km/h or more below the approach speed	W24-a chevrons on the outside of the curve, back to back, at Table 4.2 spacing	RDM 6.2 §4.6.1, Table 4.2
Descent steeper than 5 %	W30 with the actual grade	RDM 6.2 §4.6.4
Ascent steeper than 10 %	W31 with the actual grade	RDM 6.2 §4.6.4
Sight distance below passing sight distance	P32 no overtaking with a W23 distance plate at the start, and an END plate, in each direction	RDM 6.2 §3.11
Sag with headlight sight distance short of SSD; crest short of SSD	Reported as findings (no sign proposed)	AASHTO 2011 defaults

Signs are placed at the advance distance (6 seconds of travel at the design speed, RDM 6.2 §2.4), on the left for up-chainage traffic and on the right for down-chainage traffic, at the set clearance beyond the reference edge, and moved apart where they would be closer than 0.6 V metres.

On Olkaria LOT 3 (80 km/h) the engine reports 36 findings and proposes 92 faces in about 2.5 s: five curves of R 100–110 m with a safe speed near 55 km/h, descents to 10.8 %, and three short crests (K = 10, 16 and 14).

11 Running and reviewing proposals

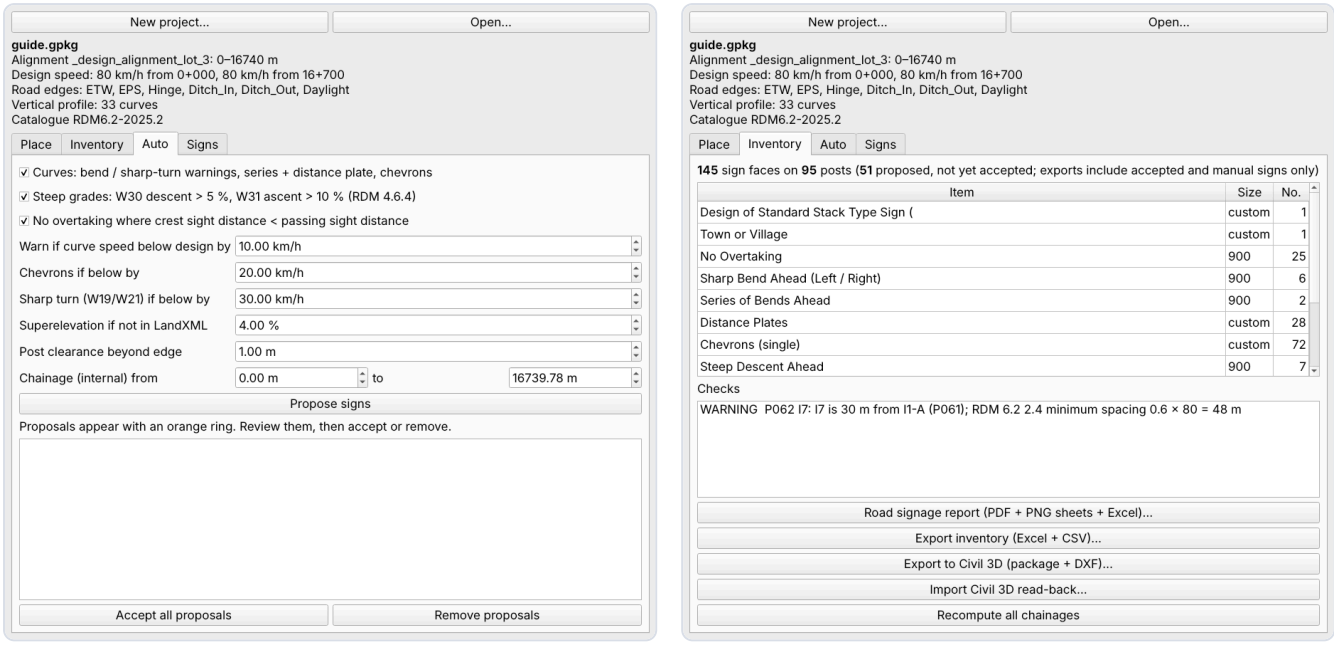


Figure 6. The Auto tab (left): rules, thresholds and chainage range. The Inventory tab (right): live counts, checks and the export buttons.

1. On the **Auto** tab, tick the rules to run and set the chainage range (the whole alignment by default).
2. **Propose signs**. Proposals are added to the sign layer with status `proposed`, drawn with an orange ring, and listed with the reason for each (for example *R100 L: curve speed 55 vs 80 km/h*).
3. Review on the map and in the report, which shows proposals too. Move, edit or delete individual proposals as needed.
4. **Accept all proposals** turns the remainder into ordinary signs. **Remove proposals** deletes all that are still proposed.

> **EXPORTS**
>
> The inventory and the Civil 3D exports include accepted and hand-placed signs only. The report includes proposals, ringed, so they can be reviewed on paper.

12 Parameters

Parameter	Default	Meaning
Warn if curve speed below design by	10 km/h	Margin that triggers a bend warning.
Chevrons if below by	20 km/h	Margin that adds chevrons.
Sharp turn if below by	30 km/h	W19/W21 instead of W20/W22.
Series gap	150 m	Warned curves closer than this are signed as a series.
Superelevation if not in LandXML	4 %	Used when the file has no superelevation for a curve.

Parameter	Default	Meaning
Steep descent / ascent	5 % / 10 %	RDM 6.2 §4.6.4 thresholds; grades shorter than 50 m are ignored.
Post clearance beyond edge	1.0 m	Within RDM's 0.6–3.0 m band.
Offset without corridor edges	5.5 m	From the centreline when the LandXML has no cross-sections.
Chainage range	whole road	Internal stations to run over.

PART 4

Outputs

The inventory for the QTO, the road signage report, and the two routes into Civil 3D drawings.

13 Inventory for the QTO

Export inventory (Excel + CSV) writes `*_Inventory.xlsx` with three named Excel Tables, ready to link into a QTO workbook, and a CSV of the faces:

Table	One row per	Main columns
RSD_Signs	sign face	Post, chainage, side, offset, edge and clearance, facing, code, description, size, legend, grade, design speed, UUIDs
RSD_Posts	post	Post ref, chainage, side, post type, foundation, number of faces
RSD_Quantities	item	Counts of faces by code and size, posts by type, foundations by type

There are no rates: the tables feed the separate QTO and Bill process.

14 Road signage report

Road signage report (PDF + PNG sheets + Excel) asks for:

- **Plan scale** for the plan layout sheets: Auto (1:2500, or 1:1000 for roads under 1.2 km), a fixed scale from 1:500 to 1:5000, or no plan sheets.
- **Prepared by, Checked by, Approved by** for the title blocks. The names are kept with the project and offered again next time.

It writes an A3 landscape PDF, a PNG per plan and long-section sheet, and an Excel workbook. Every page sits in the same frame: a border, the title with the HH logo at the top right, and a title block with Project, Alignment, Drawing, Scale, Prepared / Checked / Approved by, Date and Sheet. To use another logo, point the QGIS setting `RoadSignDesigner/report_logo` at an image file.

Plan layout sheets

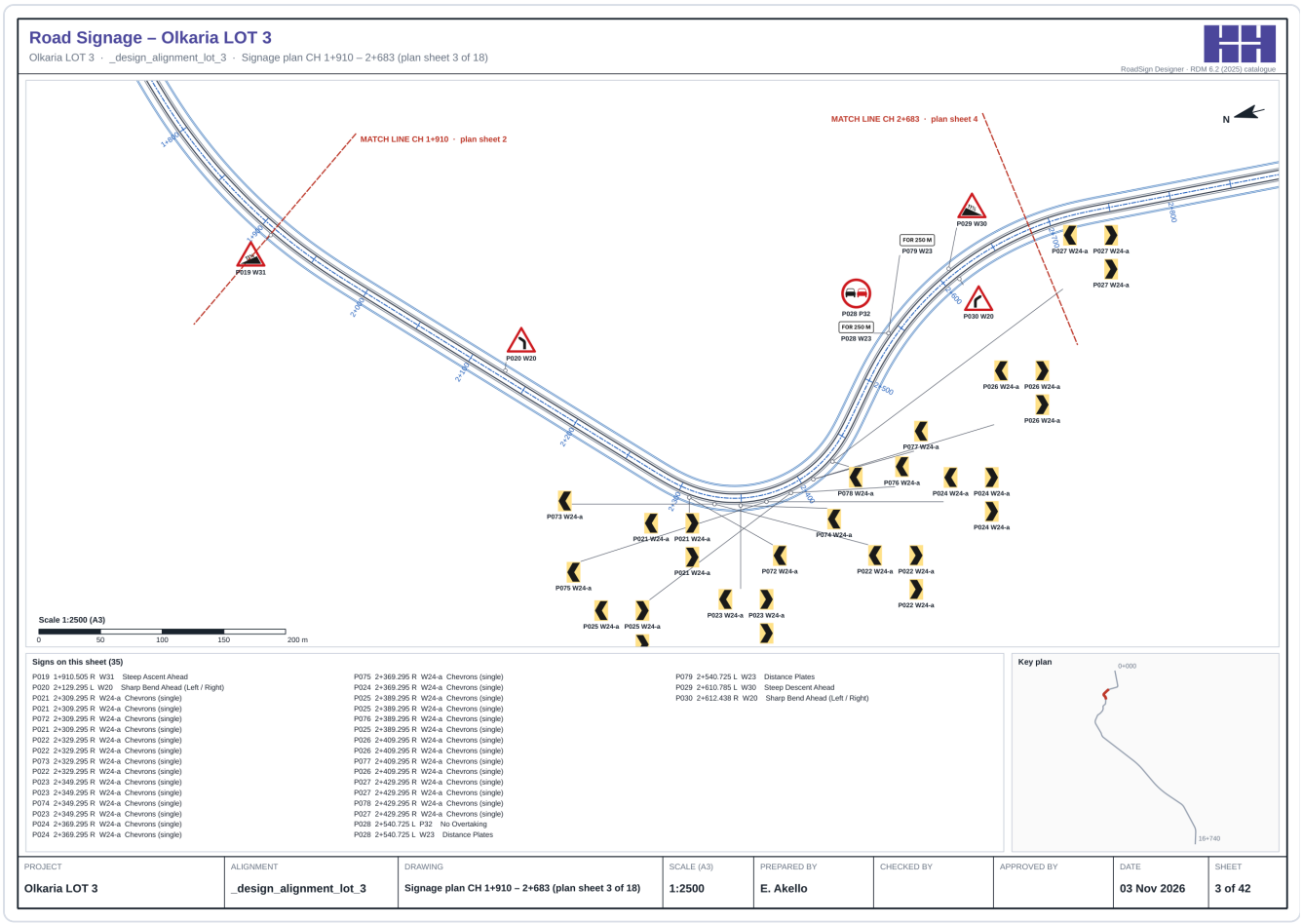


Figure 7. Plan sheet 3 of 18 for Olkaria LOT 3 at 1:2500 (CH 1+910 to 2+683): corridor edges, chainages, signs upright beside the road with leaders to their posts, match lines, north arrow, scale bar, the sheet's sign list and a key plan.

- Each sheet is turned so the road runs left to right. Sheet lengths adapt to the road: on straights a sheet covers about 950 m at 1:2500; on tight curves it is shortened so the road and the signs either side still fit. Olkaria's 16.7 km fits on 18 sheets.
- Signs are drawn upright at 10 mm, as in HH plan drawings, beside the road with a leader to the post. A sign and its plate stack; back-to-back faces sit side by side; chevrons are drawn at 70 % so a series stays readable. Symbols are moved along or away from the road until they clear one another.
- The LandXML's other alignments (side and link roads) are drawn and named.
- Match lines carry the chainage and the neighbouring sheet number.

Long-section sheets

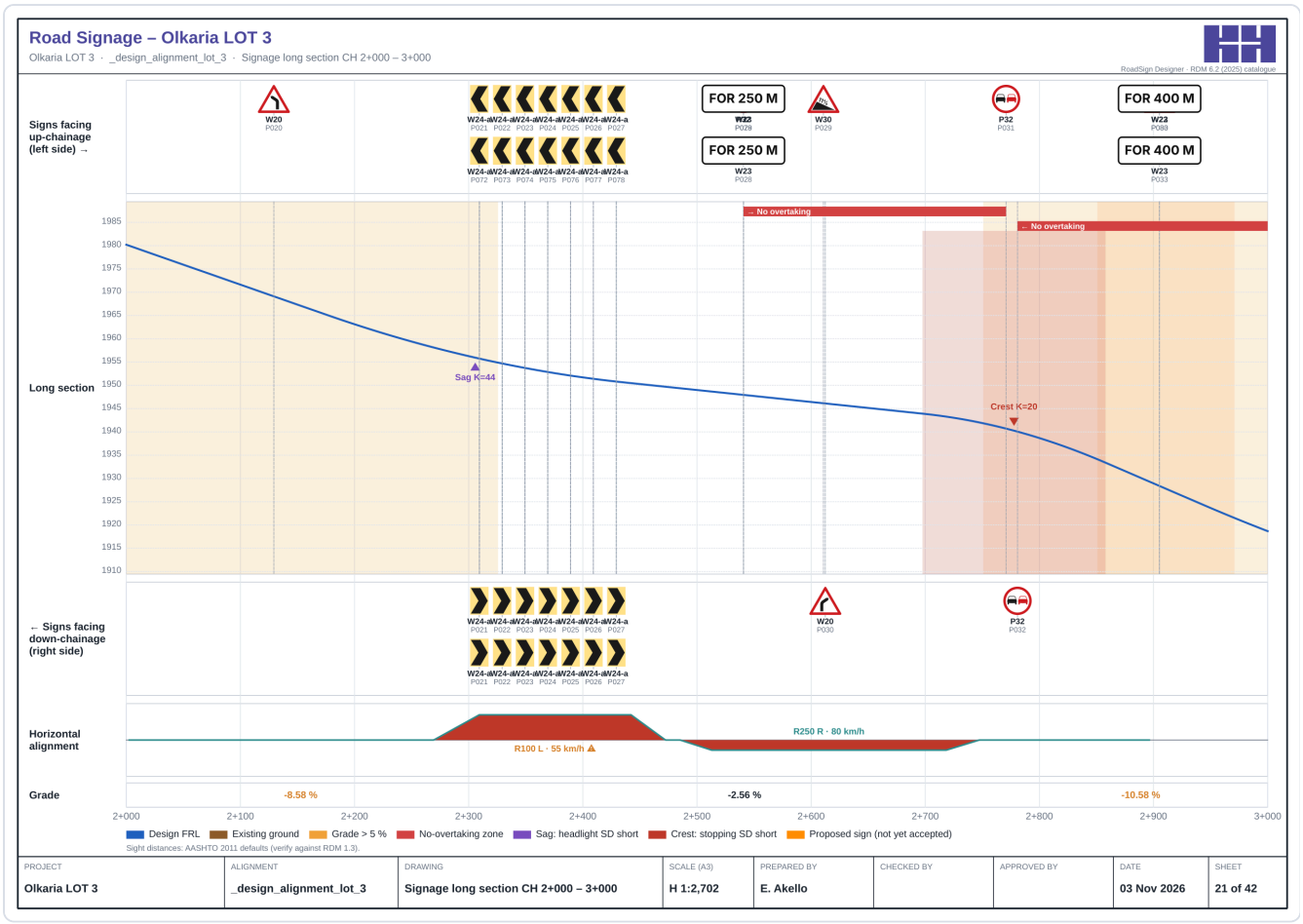


Figure 8. Long section CH 2+000 to 3+000: signs for each direction above and below the profile, steep-grade shading, no-overtaking bars, the short crest (K = 20) and its stopping-sight-distance band, the curvature band with the R 100 m curve's safe speed, and grades.

One sheet per kilometre shows the signs for each direction of travel, the design profile (and existing ground where the LandXML has it), steep grades, no-overtaking zones per direction, crests and sags with K values, sight-distance findings, the curvature band with each curve's radius and safe speed, and the grades between PVIs. Labels are staggered so that close signs, such as a chevron series, never overlap.

Schedule, findings and workbook

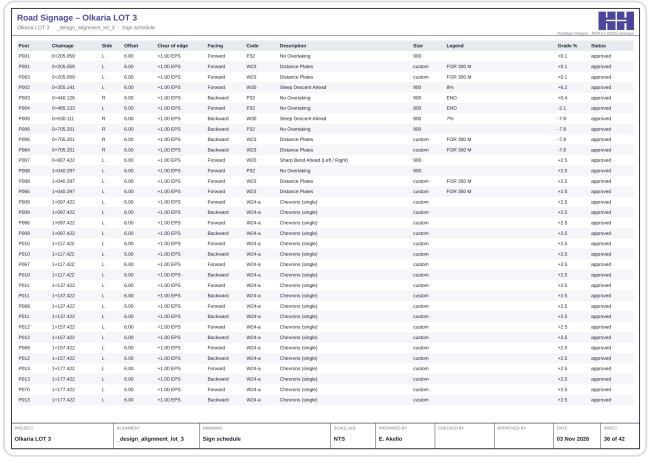


Figure 9. Sign schedule page.

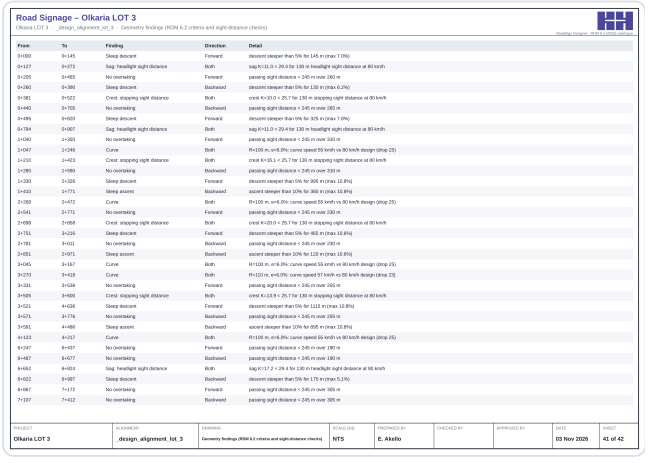


Figure 10. Geometry findings page.

The Excel workbook has the sheets Signs (with proposals and the reason for each), Findings, Horizontal curves (R, e, f, safe speed against design speed), Vertical curves (K, A, SSD) and Basis (the criteria used).

15 DXF for Civil 3D drawings

Export to Civil 3D (package + DXF) asks for the plan scale (default 1:2500) and writes the placement package (`*_placement.json`) and a DXF of the placed signs.

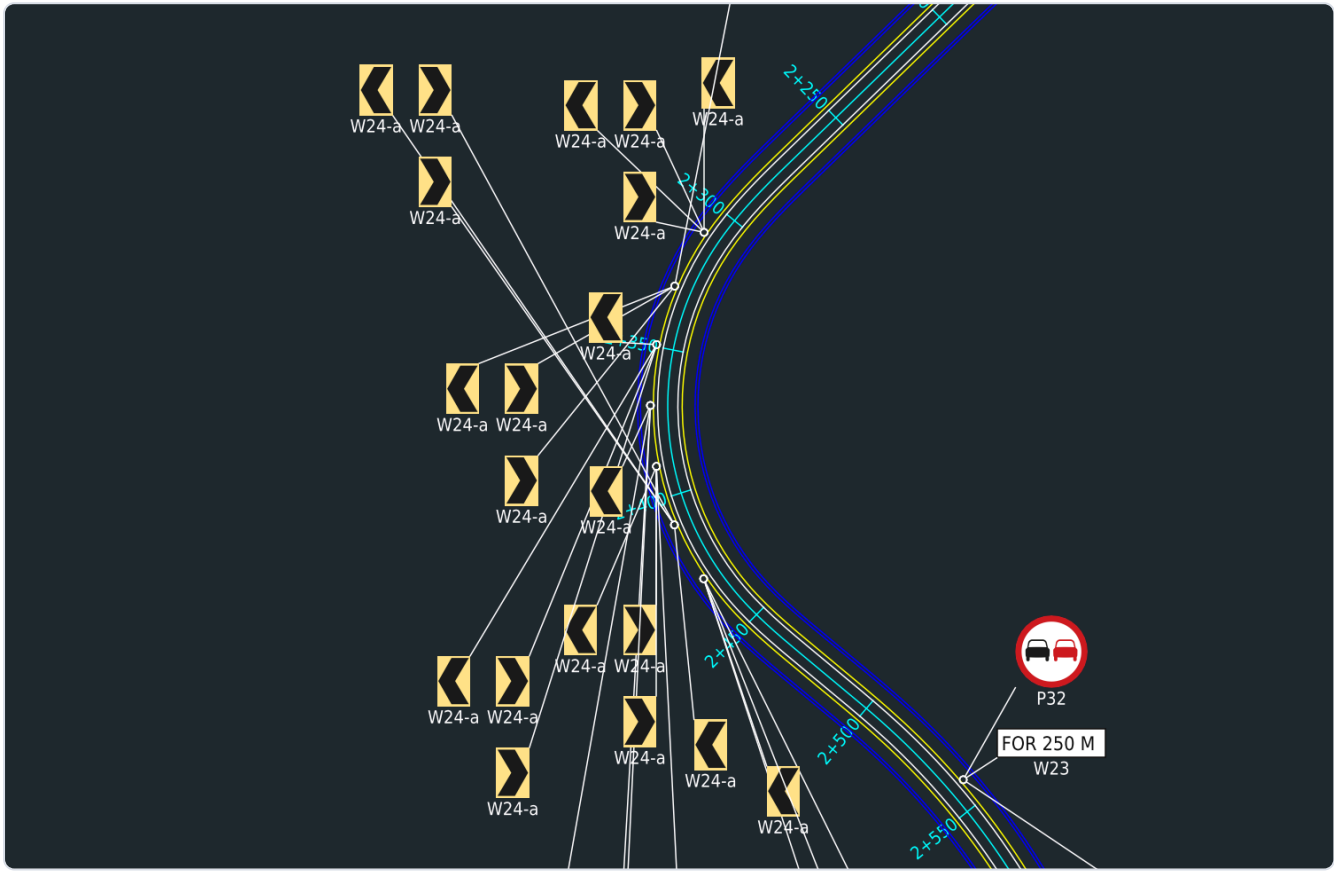


Figure 11. The exported DXF at the Olkaria chevron curve, 1:2500: each sign a block at its post with a leader to the upright face; chainages read the right way up; corridor edges and the alignment on non-plotting reference layers.

- **Size.** Symbols print at 10 mm, matching HH drawings: 25 m in model space at 1:2500, 10 m at 1:1000. Codes, legends and chainages print at 1.8 mm.
- **One block per sign, at the post.** Each sign is a block `RSDC_<code>_<id>` inserted at the post's true coordinates. Inside it are a post marker, a leader, the sign face upright (the RDM artwork block) and the code underneath. Moving, copying or reading back acts on the post.
- **Text signs.** Directional signs and plates get their own block with the real legend (colours, arrows, place names, route numbers, distances) at 0.6 of the symbol size.
- **Attributes and identity.** Every sign block carries `SIGN_CODE` , `LEGEND` , `DESCRIPTION` , `SIZE` , `CHAINAGE` , `OFFSET` , `SIDE` , `FACING` , `POST_REF` , `FACE_UUID` and `INSTALL_UUID` , with the UUIDs repeated in XDATA (application `RSD`).
- **Reference geometry** (non-plotting): the signed alignment with chainages, every other alignment in the LandXML, the corridor edges by code, and a coordinate check point.

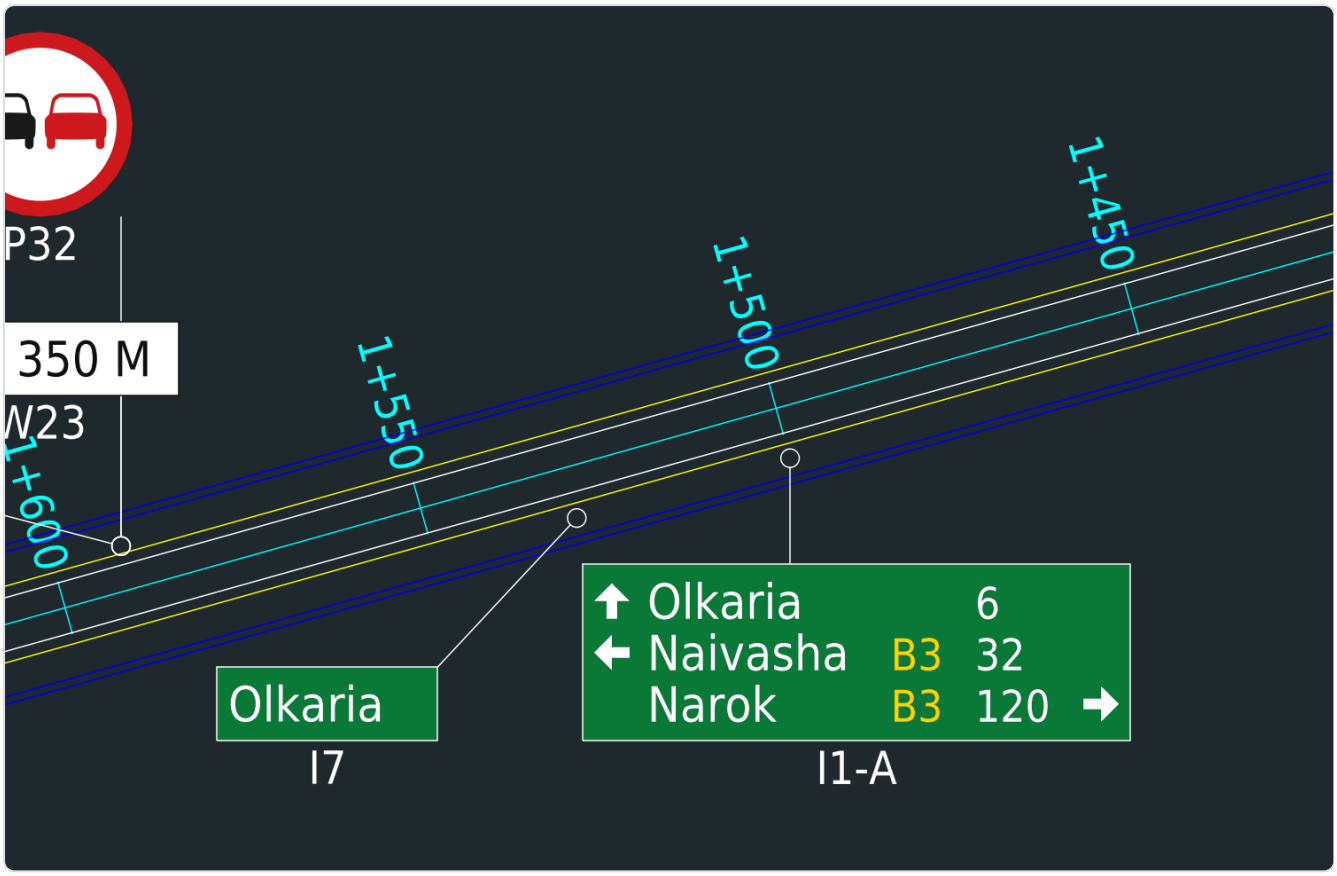


Figure 12. A directional sign and a place-name sign in the DXF, with their legends as real text.

16 Copy and paste at the true coordinates

The DXF is in the LandXML's own world coordinates: easting and northing in metres, with no local origin, shift, scale or rotation. The header records metres as the unit, angles counter-clockwise from east, and custom variables `RSD_CRS`, `RSD_MAP_ZONE`, `RSD_SOURCE_LANDXML`, `RSD_ALIGNMENT` and `RSD_COORDINATES`. Extents are set, so Zoom Extents frames the road.

1. Open `*_placement.dxf` in Civil 3D.
2. Select the signs: `QSELECT` → Layer = `RSD-SIGN`.
3. `COPYBASE` with base point `0,0,0`.
4. In the project drawing, run **`PASTEORIG`**. The signs land at their true coordinates.

Alternatively, `INSERT` or `XREF` the whole DXF at `0,0,0`, scale 1, rotation 0. The project drawing must be in metres, use the same coordinate system and be in the World UCS.

> **CHECK POINT**
> The layer `RSD-REF-NOTES` holds a cross at the start of the alignment labelled with its easting and northing. Paste it once to confirm it falls on the Civil 3D alignment, then delete it.

17 Dynamo link (parked)

Dynamo scripts that place native blocks from the placement package and read moves back into QGIS are kept in the repository (`dynamo/`), but this route is parked for now. Use the georeferenced DXF (section 16) to bring signs into Civil 3D.

PART 5

Technical manual

How the data is stored, what is read from the LandXML, the geometry and design formulas, the catalogue, the plan-layout method, and how the outputs were verified.

18 Data model

A sign project is one GeoPackage:

Table	Contents
rsd_signs	Point per sign face. User fields and derived fields (Appendix A).
rsd_meta	Key/value: LandXML path (relative), alignment, project name, schema, CRS of the LandXML, layer CRS, Prepared / Checked / Approved by.
rsd_alignment, rsd_chainage	Alignment line and chainage ticks.
rsd_edges	Corridor edges as lines, split where sections are missing.
rsd_vcurves	Vertical curve PVIs with kind, length, K, A and grades.

Faces on one post share an `install_uuid`; each face has its own `face_uuid`. The UUIDs follow a sign through the inventory, the package, the DXF and any read-back, so the same sign is recognised everywhere. Derived fields are recomputed from the alignment when a sign moves or when a field that affects them changes; they are never typed.

19 What is read from the LandXML

Element	Used for
CoordinateSystem	EPSG code and name: layer CRS, DXF header.
Alignment / CoordGeom	Line, Curve (arc) and clothoid Spiral elements; coordinates are northing first, as Civil 3D writes them.
StaEquation	Displayed chainage versus internal station.
Feature SpeedStation, Roadway/DesignSpeed	Design speed by chainage: sizes, advance distances, sight distances.
Superelevation / FullSuperelev	e for curve speeds.

Element	Used for
Profile / ProfAlign	PVIs and parabolic curves: grades, K, crests, sags, sight distance.
Profile / ProfSurf	Existing ground on the long sections.
CrossSects / DesignCrossSectSurf	Coded points (ETW, EPS, Hinge, Ditch_In, Ditch_Out, Daylight) become road edges. Pavement-layer codes are ignored.

The file is streamed (`iterparse`), and TIN surfaces are skipped, so large exports load quickly: Olkaria's 59 MB file opens in about 1.3 s.

20 Geometry methods

- **Point at station and offset.** Each element gives its point and heading in closed form (lines, arcs) or by numerical integration of curvature (clothoids). Positive offsets are to the right of increasing chainage.
- **Station and offset of a point.** The alignment is densified, the nearest segment found, and the result refined on the element itself.
- **Road edges.** For each code and side, the section offsets are interpolated along the road. An edge is treated as absent across a gap longer than three times its usual section spacing (at least 30 m), so a daylight line modelled only in cut is not bridged across fills.
- **Corridor clamp.** The post is held 0.6–3.0 m beyond the reference edge; the outer corridor limit (ditch or daylight) applies only where the verge is actually modelled.
- **Grade at a sign.** From the design profile, including inside vertical curves.

21 Design criteria and formulas

Item	Method	Source
Curve safe speed	$V^2 = 127\ R\ (e + f_{max}(V))$, solved by iteration	f_{max}: AASHTO 2011 Table 3-7 (default)
Advance distance	6 s of travel at the design speed, rounded up to 10 m (130 m at 80 km/h)	RDM 6.2 §2.4
Minimum spacing	0.6 V metres	RDM 6.2 §2.4
Lateral position	0.6–3.0 m beyond the carriageway or paved-shoulder edge	RDM 6.2 §2.4
Chevron spacing	By radius, on the curve and before it	RDM 6.2 Table 4.2
Steep grades	Descent > 5 %, ascent > 10 %	RDM 6.2 §4.6.4
Sight distance on crests	Running scan of the profile: the farthest point visible from an eye 1.08 m high, compared with SSD (object 0.60 m) and PSD (object 1.08 m)	AASHTO 2011 SSD and PSD tables (default)
Sags	Headlight sight distance (0.6 m, 1° up-spread) compared with SSD	AASHTO 2011 (default)
Sign sizes	By design speed for Stop, Give Way, warning and circular regulatory signs	RDM 6.2 Tables 3.1, 3.3, 4.1

TO CONFIRM

RDM 6.2 does not give side-friction, stopping or passing sight distance values; RDM Part 1.3 governs them. Until they are checked against Part 1.3 the AASHTO 2011 values are used and labelled as defaults in the report.

22 The sign catalogue



Figure 13. A selection of the 245 signs in the RDM 6.2 catalogue, drawn from the manual's own figures.

- **Source.** `tools/extract_rdm_signs.py` reads RDM 6.2 (2025), matches each figure caption to its framed artwork, keeps the sign and drops dimension lines and notes, and writes an SVG icon and a CAD block per sign (`RSD_<code>` , colour hatches). `tools/build_catalogue.py` adds sizes, legend types and size rules.
- **Coverage.** R, M, P, W, I, E, PD, IM and RW signs, 245 in all. Signs used both ways round have a mirrored version (`_M`). W24 chevrons are split into W24-a to W24-d.
- **Numbering clashes in the manual**, resolved as follows: Fig 7.1 captioned P1 becomes PD1; I10 used twice, so County Boundary becomes I10B; IM4 used twice, so the expressway symbol becomes IM4S.
- **Other notes.** Table 5.1 lists UC under both green and white faces (Table 2.6 is followed). Table 4.1's siting distances (1000, 900 and 600 m) look implausible, so the §2.4 six-second rule is used.
- **Coding.** RSD uses RDM 6.2 codes, and HH drawings will carry them (agreed November 2026). The Ministry's standard drawings WSCM/84/2/01 (2024) number signs differently.

23 Plan layout of symbols and sheets

At drawing scale a sign symbol is far bigger than its post (25 m at 1:2500), so drawings show it beside the road with a leader. The same method places symbols in the DXF and on the plan sheets:

1. Faces are grouped by post. Faces with the same facing stack top to bottom (sign over plate); back-to-back faces sit side by side. Each face's box includes room for its label.
2. Groups are taken in chainage order. Each is tried at candidate positions just outside its post, away from the road, then slid along the road (ahead first) and only then moved further out, until its box clears every

box already placed.

3. The leader runs from the post to the nearest point of the group's box.

Plan sheets are cut so the road fits: starting from the sheet's full length at the chosen scale, the window is shortened by 10 % at a time until the road, turned to run along the sheet, fits the viewport with room for symbols either side. A short remainder at the end of the road is merged into the last sheet.

24 Verification

Check	Data	Result
Alignment geometry	Olkaria LOT 3, Nessuit–Salgaa (4 alignments, 26 spirals)	Element ends reproduce the LandXML coordinates. A 0.096° kink at Rongai–Salgaa 0+536 is in the source file.
DXF georeferencing	Olkaria LOT 3, 92 signs	Reference alignment starts exactly at the LandXML's first point (E 204708.247, N 9898762.709). Every sign back-computes to its chainage and offset with 0.000 mm difference.
Paste into a project drawing	Olkaria LOT 3 DXF into the project DWG (Civil 3D)	COPYBASE 0,0 and PASTEORIG: signs and plates landed on the alignment at their chainages.
DXF posts after the callout change	Automated test	Every sign block's insertion point equals its post coordinates; read-back finds the sign's own block name.
Auto-placement	Olkaria LOT 3, 80 km/h	36 findings, 92 proposed faces, no check errors. Nessuit–Salgaa (70 km/h): no proposals, as its geometry meets the design speed.
Report	Olkaria LOT 3	18 plan sheets at 1:2500, 17 long sections, schedule and findings; about 40 pages in about 8 s.
Automated tests	Synthetic roads	37 core tests, 27 headless QGIS tests.

25 Limitations and roadmap

- **Civil 3D.** The DXF paste works; the Dynamo link is parked.
- **HH drawing conventions.** Symbol sizes now match the 25 m blocks in current HH drawings at 1:2500. Block names, layers and annotation style should be matched to HH's once a DXF export of a current drawing is available.
- **Quantities.** Post and foundation types for the QTO are still to be agreed. Sign numbering follows RDM 6.2.
- **Corridor restriction.** Implemented as RDM's 0.6–3.0 m band inside the ditch or daylight line. A road-reserve boundary is not used yet.
- **Junctions.** Junction signing (RDM Table A1-1) is not automated; place those signs by hand.
- **Sight distances** use AASHTO 2011 defaults until checked against RDM Part 1.3.

26 Troubleshooting

Symptom	Cause and fix
Export says to fix errors first	A check is at error level (for example a post inside the travelled way). Fix it on the Inventory tab's list.
No road edges or "Offset from" list empty	The LandXML has no coded corridor sections. Export the corridor sections from Civil 3D, or use offsets from the centreline.
Signs pasted in the wrong place	The project drawing is in another coordinate system, in millimetres, or in a user UCS. Check the RSD CHECK POINT against the alignment.
Export libraries missing	The plugin's `ext_libs` folder is missing. Reinstall from the ZIP.
Old project lacks newer fields	Fields are added automatically on open. If the GeoPackage is read-only, copy it somewhere writable.

Appendix A · Fields and layers

Sign layer fields (rsd_signs)

Field	Kind	Meaning
face_uuid, install_uuid	identity	Face and post identities, kept across all exchanges.
post_ref	user	Post number (P001…).
sign_code, size, size_auto, mirror	user	RDM code, size (mm or custom), Auto size, mirrored version.
facing, facing_auto, skew	user	Forward / Backward, Auto facing, skew in degrees.
legend, road_class	user	Legend text and colour scheme.
post_type, foundation, remarks	user	For the QTO.
status, rule, reason	engine	`proposed`, `approved` (accepted proposal) or empty for hand-placed; the rule and the reason.
chainage, chainage_m, station_int	derived	Displayed chainage (text and metres) and internal station.
offset_m, side, face_az	derived	Offset from the centreline, L/R, face azimuth.
edge_ref, edge_clear	derived	Reference edge and clearance beyond it.
grade_pct, speed_kmh	derived	Design grade and speed at the sign.
description, face_svg, stack	derived	Sign name, generated face file, position in the post's stack.

DXF layers

Layer	Plots	Contents
RSD-SIGN	yes	Sign blocks (one per face, inserted at the post).
RSD-SIGN-TEXT	yes	Attribute text (code, legend).
RSD-SIGN-LEADER	yes	Post markers and leaders.
RSD-ALIGN	no	Signed alignment and chainages.
RSD-ALIGN-OTHER	no	Other alignments in the LandXML, named.
RSD-EDGE-<code>	no	Corridor edges by code.
RSD-REF-NOTES	no	Coordinate check point.